

O(Alpha)

Quant in your pocket.

An automated quantitative trading platform that turns risk appetite into systematic, emotion-free execution.

NVDA +2.8% AAPL +0.9% BTC +1.4% SPY +0.6% MSFT +1.0% ETH +2.1% QQQ +1.1% AVGO +1.8%



01

Configure your agent

Set a risk profile: the agent maps it to leverage, stop-loss and rebalance parameters.



02

Systematic strategies

Run rigorously tested strategies from research to cost-stress gates, including HMM regime routing.



03

Track everything

Monitor P&L, risk controls, positions and the complete execution audit trail.

SHARPE RATIO

2.34

MAX DRAWDOWN

-12.3%

ANNUAL RETURN

18.7%

O(Alpha) / OVERVIEW

AGENT STATUS + OPTIMISING

REGIME · VOLATILE

CURRENT P&L (24H)

+\$12,450.89

ILLUSTRATIVE · NOT INVESTMENT ADVICE

RISK TOLERANCE

High

VOLATILITY CAP

30%

LEVERAGE

PERFORMANCE

AAPL · 1D

SHARPE

1.84

MAX DD

-8.4%

KALMAN

MARKET REGIME

Bull 74%
Volatile 50%
Bear 24%

MOTIVATION

Retail investors need quant-grade automation without deep infrastructure work. O(Alpha) turns strategy research into accessible, systematic execution.

The problem

Research, execution and account state can drift apart when validation, background workers and persistence are loosely connected.

Our solution

A gated research-to-execution pipeline with explicit lifecycle states, normalized data and a reconstructable audit trail.

USER STORIES

BEGINNER

Choose a strategy that matches a saved risk profile.

RESEARCHER

Inspect repeatable validation evidence before launch.

PORTFOLIO

Start, observe and stop one strategy agent safely.

MAINTAINER

Trace each run from target weights to persisted fills.

ENGINEERED FEATURES

VERIFIABLE

Onboarding & Catalog

Risk profile, compatible strategy and accepted backtest stay aligned.

Execution Router

Deterministic order keys; reductions are processed before buys.

Lifecycle Resume

Run state, heartbeat and runtime HMM state survive restarts.

Validation Gate

No-lookahead, walk-forward, DSR/PBO and cost stress; missing evidence fails closed.

Atomic Ledger

Orders, fills, cash and positions update inside DB transactions.

Observable Dashboard

Snapshots, positions, fills and alerts come from persisted state.

BACKEND TESTS

50

FRONTEND TESTS

3

CI WORKFLOWS

3

BACKEND

Go API, Gin, gorm, secure auth and Alpaca SDK.

DATA

PostgreSQL, TimescaleDB, Redis pub/sub and Alpaca API.

FRONTEND

Next.js, TypeScript, Recharts and Vercel.

SOFTWARE ENGINEERING

ARCH

Layered UI, API, domain and data boundaries with normalized account and ledger records.

SAFETY

Invalid risk, missing artifacts and unknown promotion evidence fail closed.

LIFE

One active run per user, explicit states, heartbeats and restart recovery.

PROOF

Claims trace to tests, CI gates and committed research artifacts.

TEST PYRAMID

UNIT + VALIDATION

50 Go test files cover APIs, agents, backtests, ML parity and promotion gates.

INTEGRATION + UI

DB agent repository integration and Playwright settings-lock smoke coverage.

SYSTEM ARCHITECTURE



Next.js Client

Onboarding, control portfolio views



Go API Server

Auth, validation lifecycle commands



Domain Engine

Catalog, backtest portfolio orchestrator



PostgreSQL

Orders, fills, ledger runs, bars, alerts



Alpaca Markets

REST + WebSocket market integration

Go

Gin

Next.js

React

TypeScript

PostgreSQL

TimescaleDB

Redis

Docker

Vercel

Alpaca

Tailwind